# DATABASE MANAGEMENT SYSTEMS
# BECS 22243

## PREPARATION 01

EC/2021/005 – Thamara G.P.D.

Create tables

```
SQL> CREATE TABLE Part_Table(
  2  PNO CHAR(2)NOT NULL PRIMARY KEY,
  3  PNAME VARCHAR(20),
  4  PRICE NUMBER,
  5  COLOR VARCHAR(10),
  6  WEIGHT NUMBER,
  7  CITY VARCHAR(15)
  8  );

Table created.

SQL> CREATE TABLE Projec_Table
  2  (JNO CHAR(2)NOT NULL PRIMARY KEY,
  3  PNAME VARCHAR(20),
  4  CITY VARCHAR(15));

Table created.

SQL> CREATE TABLE Supplier_Table(
  2  SNO CHAR(2)NOT NULL PRIMARY KEY,
  3  SNAME VARCHAR(20),
  4  STATYS NUMBER'
  5  CITY VARCHAR(20));
STATYS NUMBER'
             *
ERROR at line 4:
ORA-01756: quoted string not properly terminated


SQL>
SQL> CREATE TABLE Supplier_Table(
  2  SNO CHAR(2)NOT NULL PRIMARY KEY,
  3  SNAME VARCHAR(20),
  4  STATYS NUMBER,
  5  CITY VARCHAR(20));

Table created.
```

```
SQL> CREATE TABLE SPJ_Table(
  2  SNO CHAR(2),
  3  PNO CHAR(2),
  4  JNO CHAR(2),
  5  QTY NUMBER(5),
  6  PRIMARY KEY (SNO,PNO,JNO),
  7  FOREIGN KEY (SNO) REFERENCES  Supplier_Table(SNO),
  8  FOREIGN KEY (PNO) REFERENCES  Part_Table(PNO),
  9  FOREIGN KEY (JNO) REFERENCES  Projec_Table(JNO)
 10  );

Table created.
```

Insert values into tables

```
SQL> INSERT ALL
  2  INTO Part_Table VALUES ('P1','NUT',20.00,'RED',12,'COLOMBO')
  3  INTO Part_Table VALUES ('P2','BOLT',14.00,'GREEN',17,'KANDY')
  4  INTO Part_Table VALUES ('P3','SCREW',14.00,'BLUE',17,'MATARA')
  5  INTO Part_Table VALUES ('P4','SCREW',11.00,'RED',14,'COLOMBO')
  6  INTO Part_Table VALUES ('P5','CAM',150.00,'BLUE',12,'KANDY')
  7  INTO Part_Table VALUES ('P6','COG',12.00,'RED',19,'COLOMBO')
  8  SELECT * FROM DUAL;

6 rows created.

SQL> COMMIT;

Commit complete.

SQL> INSERT ALL
  2  INTO Projec_Table VALUES ('J1','SORTER','KANDY')
  3  INTO Projec_Table VALUES ('J2','PUNCH','MATARA')
  4  INTO Projec_Table VALUES ('J3','READER','JAFFNA')
  5  INTO Projec_Table VALUES ('J4','CONSOLE','JAFFNA')
  6  INTO Projec_Table VALUES ('J5','COLLATOR','COLOMBO')
  7  INTO Projec_Table VALUES ('J6','VIDEO','GALLE')
  8  INTO Projec_Table VALUES ('J7','TAPE','COLOMBO')
  9  SELECT *FROM DUAL;

7 rows created.

SQL> COMMIT;

Commit complete.

SQL> INSERT ALL
  2  INTO Supplier_Table VALUES ('S1','SAMAN',20,'COLOMBO')
  3  INTO Supplier_Table VALUES ('S2','JAGATH',10,'KANDY')
  4  INTO Supplier_Table VALUES ('S3','ANIL',30,'KANDY')
  5  INTO Supplier_Table VALUES ('S4','GAMINI',20,'COLOMBO')
  6  INTO Supplier_Table VALUES ('S5','BALA',30,'JAFFNA')
  7  SELECT *FROM DUAL;

5 rows created.

SQL> COMMIT;

Commit complete.
```

```
SQL> INSERT ALL
  2  INTO SPJ_Table VALUES ('S1','P1','J1',200)
  3  INTO SPJ_Table VALUES ('S1','P2','J4',700)
  4  INTO SPJ_Table VALUES ('S2','P3','J2',200)
  5  INTO SPJ_Table VALUES ('S2','P3','J3',200)
  6  INTO SPJ_Table VALUES ('S2','P3','J4',500)
  7  INTO SPJ_Table VALUES ('S2','P3','J5',600)
  8  INTO SPJ_Table VALUES ('S2','P3','J6',400)
  9  INTO SPJ_Table VALUES ('S2','P3','J7',800)
 10  INTO SPJ_Table VALUES ('S2','P5','J2',100)
 11  INTO SPJ_Table VALUES ('S3','P3','J3',200)
 12  INTO SPJ_Table VALUES ('S3','P4','J4',500)
 13  INTO SPJ_Table VALUES ('S4','P2','J1',300)
 14  INTO SPJ_Table VALUES ('S4','P2','J4',300)
 15  INTO SPJ_Table VALUES ('S5','P1','J2',1000)
 16  INTO SPJ_Table VALUES ('S5','P2','J2',200)
 17  INTO SPJ_Table VALUES ('S5','P2','J4',100)
 18  INTO SPJ_Table VALUES ('S5','P3','J4',1200)
 19  INTO SPJ_Table VALUES ('S5','P4','J4',800)
 20  INTO SPJ_Table VALUES ('S5','P5','J4',400)
 21  INTO SPJ_Table VALUES ('S5','P5','J5',500)
 22  INTO SPJ_Table VALUES ('S5','P5','J7',100)
 23  INTO SPJ_Table VALUES ('S5','P6','J2',200)
 24  SELECT *FROM DUAL;

22 rows created.
```

```
SQL> SELECT *FROM Part_Table;

PN PNAME                    PRICE COLOR          WEIGHT CITY
-- -------------------- ---------- ---------- ---------- ---------------
P1 NUT                          20 RED                12 COLOMBO
P2 BOLT                         14 GREEN              17 KANDY
P3 SCREW                        14 BLUE               17 MATARA
P4 SCREW                        11 RED                14 COLOMBO
P5 CAM                         150 BLUE               12 KANDY
P6 COG                          12 RED                19 COLOMBO

6 rows selected.
```

```
SQL> SELECT *FROM Projec_Table;

JN PNAME                 CITY
-- -------------------- --------------
J1 SORTER                KANDY
J2 PUNCH                 MATARA
J3 READER                JAFFNA
J4 CONSOLE               JAFFNA
J5 COLLATOR              COLOMBO
J6 VIDEO                 GALLE
J7 TAPE                  COLOMBO

7 rows selected.
```

```
SQL> SELECT *FROM Supplier_Table;

SN SNAME                   STATYS CITY
-- -------------------- ---------- --------------------
S1 SAMAN                        20 COLOMBO
S2 JAGATH                       10 KANDY
S3 ANIL                         30 KANDY
S4 GAMINI                       20 COLOMBO
S5 BALA                         30 JAFFNA
```

```
SQL> SELECT *FROM SPJ_Table;

SN PN JN        QTY
-- -- -- ----------
S1 P1 J1        200
S1 P2 J4        700
S2 P3 J2        200
S2 P3 J3        200
S2 P3 J4        500
S2 P3 J5        600
S2 P3 J6        400
S2 P3 J7        800
S2 P5 J2        100
S3 P3 J3        200
S3 P4 J4        500

SN PN JN        QTY
-- -- -- ----------
S4 P2 J1        300
S4 P2 J4        300
S5 P1 J2       1000
S5 P2 J2        200
S5 P2 J4        100
S5 P3 J4       1200
S5 P4 J4        800
S5 P5 J4        400
S5 P5 J5        500
S5 P5 J7        100
S5 P6 J2        200

22 rows selected.
```

01. Get part numbers for all parts.

```
SQL> SELECT PNO FROM Part_Table;

PN
--
P1
P2
P3
P4
P5
P6

6 rows selected.
```

02. Get the supplier number, name, status and location of all suppliers.

```
SQL> SELECT * FROM Supplier_Table;

SN SNAME                    STATYS CITY
-- -------------------- --------- --------------------
S1 SAMAN                         20 COLOMBO
S2 JAGATH                        10 KANDY
S3 ANIL                          30 KANDY
S4 GAMINI                        20 COLOMBO
S5 BALA                          30 JAFFNA

SQL> SELECT SNO, SNSME, STASTUS, CITY FROM Supplier_Table;
```

03. Get the part numbers for all parts currently being supplied.

```
SQL> SELECT DISTINCT PNO FROM SPJ_Table;

PN
--
P4
P1
P2
P3
P6
P5

6 rows selected.
```

04. Get supplier numbers for suppliers in Kandy with >20 status value.

```
SQL> SELECT SNO FROM Supplier_TabLE WHERE CITY = 'KANDY' AND STATYS >20;

SN
--
S3
```

05. Get suppliers whose location is known, i.e. not null.

```
SQL> SELECT * FROM Supplier_Table WHERE CITY IS NOT NULL;

SN SNAME                    STATYS CITY
-- -------------------- ---------- --------------------
S1 SAMAN                        20 COLOMBO
S2 JAGATH                       10 KANDY
S3 ANIL                         30 KANDY
S4 GAMINI                       20 COLOMBO
S5 BALA                         30 JAFFNA
```

06. Find the parts which are located in Colombo and have weight greater than 15.

```
SQL> SELECT * FROM Part_Table WHERE CITY = 'COLOMBO' AND WEIGHT > 15;

PN PNAME                    PRICE COLOR         WEIGHT CITY
-- -------------------- ---------- ---------- ---------- ---------------
P6 COG                          12 RED                19 COLOMBO
```

07. Find the suppliers who have status values between 10 and 20.

```
SQL> SELECT SNO, SNAME FROM Supplier_Table WHERE STATYS >=10 AND STATYS <=20
;

SN SNAME
-- --------------------
S1 SAMAN
S2 JAGATH
S4 GAMINI
```

08. Get the details of suppliers who are either located in Colombo or in Jaffna.

```
SQL> SELECT * FROM Supplier_Table WHERE CITY IN ('COLOMBO' , 'JAFFNA');

SN SNAME                    STATYS CITY
-- -------------------- ---------- --------------------
S1 SAMAN                        20 COLOMBO
S4 GAMINI                       20 COLOMBO
S5 BALA                         30 JAFFNA
```

09. For each part in Colombo get the name and its weight in Kilograms.

```
SQL> SELECT PNAME, WEIGHT/1000 AS KG_WEIGHT FROM Part_Table WHERE CITY = 'COLOMBO';

PNAME                KG_WEIGHT
-------------------- ---------
NUT                       .012
SCREW                     .014
COG                       .019
```

10. Get the supplier number and the status for suppliers in Kandy, in descending order of the status.

```
SQL> SELECT SNAME, STATYS FROM Supplier_Table WHERE CITY = 'KANDY' ORDER BY STATYS DESC;

SNAME                  STATYS
-------------------- ---------
ANIL                        30
JAGATH                      10
```

11. Get all suppliers whose name has ‘A’ as its second character.

```
SQL> SELECT SNAME FROM Supplier_Table WHERE SNAME LIKE '_A%';

SNAME
--------------------
SAMAN
JAGATH
GAMINI
BALA
```

12. Retrieve the part names that start with letter C.

```
SQL> SELECT PNAME FROM Part_Table WHERE PNAME LIKE 'C%';

PNAME
--------------------
CAM
COG
```

13. Retrieve the surnames of suppliers whose name begins with any vowels.

```
SQL> SELECT SNAME FROM Supplier_Table WHERE REGEXP_LIKE(SNAME,'^[A,E,I,O,U]');

SNAME
--------------------
ANIL
```

14. Retrieve the surnames of suppliers whose name begin with any letter A to P.

```
SQL> SELECT SNAME FROM Supplier_Table WHERE REGEXP_LIKE(SNAME,'^[A-P]');

SNAME
--------------------
JAGATH
ANIL
GAMINI
BALA
```

15. Retrieve the surnames of suppliers who have an even number as the first character of their status.

```
SQL> SELECT SNAME FROM Supplier_Table WHERE REGEXP_LIKE(STATYS,'^[2,4,6,8]');

SNAME
--------------------
SAMAN
GAMINI
```

16. Get the total number of suppliers.

```
SQL> SELECT COUNT(*) AS NUMBER_OF_SUPPLIERS FROM Supplier_Table;

NUMBER_OF_SUPPLIERS
--------------------
                   5
```

17. Get the total number of suppliers currently supplying parts.

```
SQL> SELECT COUNT(DISTINCT SNO) AS TOTAL_NUMBER_OF_SUPPLIERS FROM SPJ_Table;


TOTAL_NUMBER_OF_SUPPLIERS
-------------------------
                        5
```

18. Find the maximum and minimum weight of parts.

```
SQL> SELECT MAX(WEIGHT) AS MAX_WEIGHT,MIN(WEIGHT) AS MIN_WEIGHT FROM Part_Table;

MAX_WEIGHT MIN_WEIGHT
---------- ----------
        19         12
```

19. Find the number of shipments for part P2.

```
SQL> SELECT COUNT(*) FROM SPJ_Table WHERE PNO='P2';

  COUNT(*)
----------
         5
```

20. Find the average price of parts located in Colombo.

```
SQL> SELECT AVG(PRICE) AS AVG_PRICE FROM Part_Table WHERE CITY = 'COLOMBO';

 AVG_PRICE
---------
14.3333333
```

21. Get the average price of the parts which are located in all cities other than 'Matara'.

```
SQL> SELECT AVG(PRICE) AS AVG_PRICE_NOT_MATARA FROM Part_Table WHERE CITY <> 'MATARA';

AVG_PRICE_NOT_MATARA
--------------------
                41.4
```

22. For each part currently being supplied, get the part number and the total quantity supplied for that part.

```
SQL> SELECT PNO, SUM(QTY) FROM SPJ_Table GROUP BY PNO;

PN   SUM(QTY)
-- ----------
P4       1300
P1       1200
P2       1600
P3       4100
P6        200
P5       1100

6 rows selected.
```

23. For each supplier and each part, get the total quantity supplied of this part by this supplier

```
SQL> SELECT SNO, PNO,SUM(QTY) FROM SPJ_Table GROUP BY SNO,PNO;

SN PN   SUM(QTY)
-- -- ----------
S1 P1        200
S2 P5        100
S3 P3        200
S5 P4        800
S5 P6        200
S4 P2        600
S5 P1       1000
S2 P3       2700
S5 P2        300
S3 P4        500
S5 P5       1000

SN PN   SUM(QTY)
-- -- ----------
S5 P3       1200
S1 P2        700
```

24. For each part supplied, get the number of suppliers currently supplying that part.

```
SQL> SELECT PNO, COUNT(DISTINCT SNO) AS NUM FROM SPJ_Table GROUP BY PNO;

PN        NUM
-- ----------
P4          2
P1          2
P2          3
P3          3
P6          1
P5          2

6 rows selected.
```

25. For each part supplied, get the number of shipments for that part.

```
SQL> SELECT PNO, COUNT(*) AS NUM FROM SPJ_Table GROUP BY PNO;

PN        NUM
-- ----------
P4          2
P1          2
P2          5
P3          8
P6          1
P5          4

6 rows selected.
```

26. For each supplier and each part, get the number of projects that the supplier is currently supplying with the part.

```
SQL> SELECT SNO, PNO, COUNT(JNO) FROM SPJ_Table GROUP BY SNO, PNO;

SN PN COUNT(JNO)
-- -- ----------
S1 P1          1
S1 P2          1
S2 P3          6
S2 P5          1
S3 P3          1
S3 P4          1
S4 P2          2
S5 P1          1
S5 P2          2
S5 P3          1
S5 P4          1

SN PN COUNT(JNO)
-- -- ----------
S5 P5          3
S5 P6          1

13 rows selected.
```

27. Get the part numbers for all parts supplied in more than one shipment.

```
SQL> SELECT PNO FROM SPJ_Table GROUP BY PNO HAVING COUNT(*)>1;

PN
--
P4
P1
P2
P3
P5
```

28. Calculate the average weight of the parts located in Colombo and located in Kandy.

```
SQL> SELECT CITY, AVG(WEIGHT) FROM Part_Table GROUP BY CITY HAVING CITY IN ('COLOMBO','KANDY');

CITY           AVG(WEIGHT)
-------------- -----------
COLOMBO                 15
KANDY                 14.5
```

29. For each part, the total quantity supplied is greater than 1000, get the part number and the maximum quantity of that part supplied.

```
SQL> SELECT PNO, MAX(QTY) FROM SPJ_Table GROUP BY PNO HAVING SUM(QTY)>1000;

PN   MAX(QTY)
-- ----------
P4        800
P1       1000
P2        700
P3       1200
P5        500
```

30. For the above query, order the result in increasing quantity value.

```
SQL> SELECT PNO,MAX(QTY) FROM SPJ_Table GROUP BY PNO HAVING SUM(QTY)>1000 ORDER BY 2;

PN   MAX(QTY)
-- ----------
P5        500
P2        700
P4        800
P1       1000
P3       1200
```

31. For the above query exclude all the shipments which has quantity less than 200.

```
SQL> SELECT PNO, MAX(QTY) FROM SPJ_Table WHERE QTY>200 GROUP BY PNO HAVING SUM(QTY)>1000 ORDER BY 2;

PN   MAX(QTY)
-- ----------
P2        700
P4        800
P3       1200
```